

浙江财经学院 2019 ~ 2020 学年第一学期

《数据库原理及应用（英）》课程期末考试试卷（A 卷）

考核方式： 闭 卷

考试日期：2020 年 1 月 日

适用专业、班级：17 电子商务

题 号	一	二	三	四	五	六	七	八	九	十	总分
得 分											
评卷人											

(共 7 大题)

1. Questions (Points: 15)

- (1) What is a data model? Discuss three components of data model?(7)
- (2) What is view? What are the advantages of the view?(8)

2. For each of the following questions there are four choices marked A, B, C and D. Only one of them is correct. You should decide on the correct choice and write down its mark on the answer sheet. (Points 8)

- (1) In SQL, in order to delete an existing base table S, we use ____ statement.
- A. DELETE TABLE S B. DELETE S
- C. DROP TABLE S D. DROP S
- (2) It is ____ that ensures the data independence in database system.
- A. the schema and the external schema
- B. the schema and the internal schema
- C. the two mappings between the three level of schemas
- D. the three level architecture of schemas
- (3) If a relation schema belongs to 2NF, then it ____.
- A. may belong to 1NF B. certainly belongs to 1NF
- C. certainly belongs to 3NF D. certainly belongs to BCNF
- (4) Which one of the following is not the property of relations?
- A. Each attribute has a distinct data type. B. Each attribute has a distinct name.
- C. Order of rows has no significance. D. Order of cols has no significance.
- (5) Given a relation schema R and its attributes X, if X references the primary key Y of relation schema S, then we call ____.
- A. X is a foreign key of relation schema S B. Y is a foreign key of relation schema R
- C. X is a foreign key of relation schema R D. Y is a foreign key of relation schema S
- (6) Null represents a value for an attribute in a tuple that is ____.

A. zero B. an empty set C. a space D. currently unknown or not applicable

(7) Suppose there are 10 tuples in each of relation R , S and W , then the number of tuples in $R \times S \times W$ is _____.

A. 10 B. 30 C. 1000 D. unknown

(8) In SQL, CREATE, DROP and ALTER are _____.

A. Data query language B. Data Modification Language(DML)
C. Data Definition Language(DDL) D. Data Control Language(DCL)

3. Given the following descriptions, create an appropriate ER diagram for the management of the Library. (point:12)

In a library, a reader can borrow zero or more books; a book can be borrowed by zero or more readers; an author can write one or more books, but for each book, we only store its first author. Books should have attributes such as a unique book number, Title, publisher, publish time, price, ISBN, each book belongs to a certain category and there are a number of books for each category. Each category has a unique number and a description. Readers should have attributes such as a unique reader number, reader name, DOB(date of birth)gender(性别), address, telephone number, Email. Authors should have attributes such as A unique author number, author name, birth-year, gender, affiliation(所在单位), style. When a reader borrows a book, we must register the borrowing date, and the date due to return, when a reader returns a book, we must record the date of return..

There are several rooms in the library for list out all the books. In each room, there are a number of book shelves(书架). All the books are grouped by category, and then placed on certain book shelves. One book shelf lists the books that belong to a same category. That is to say different kinds of books were placed on different book shelves. Each room has a unique number, telephone, assistant and the number of book shelves, each book shelf has a unique code, length.

4. Calculation of relational algebra (2 points each, total 10 points).

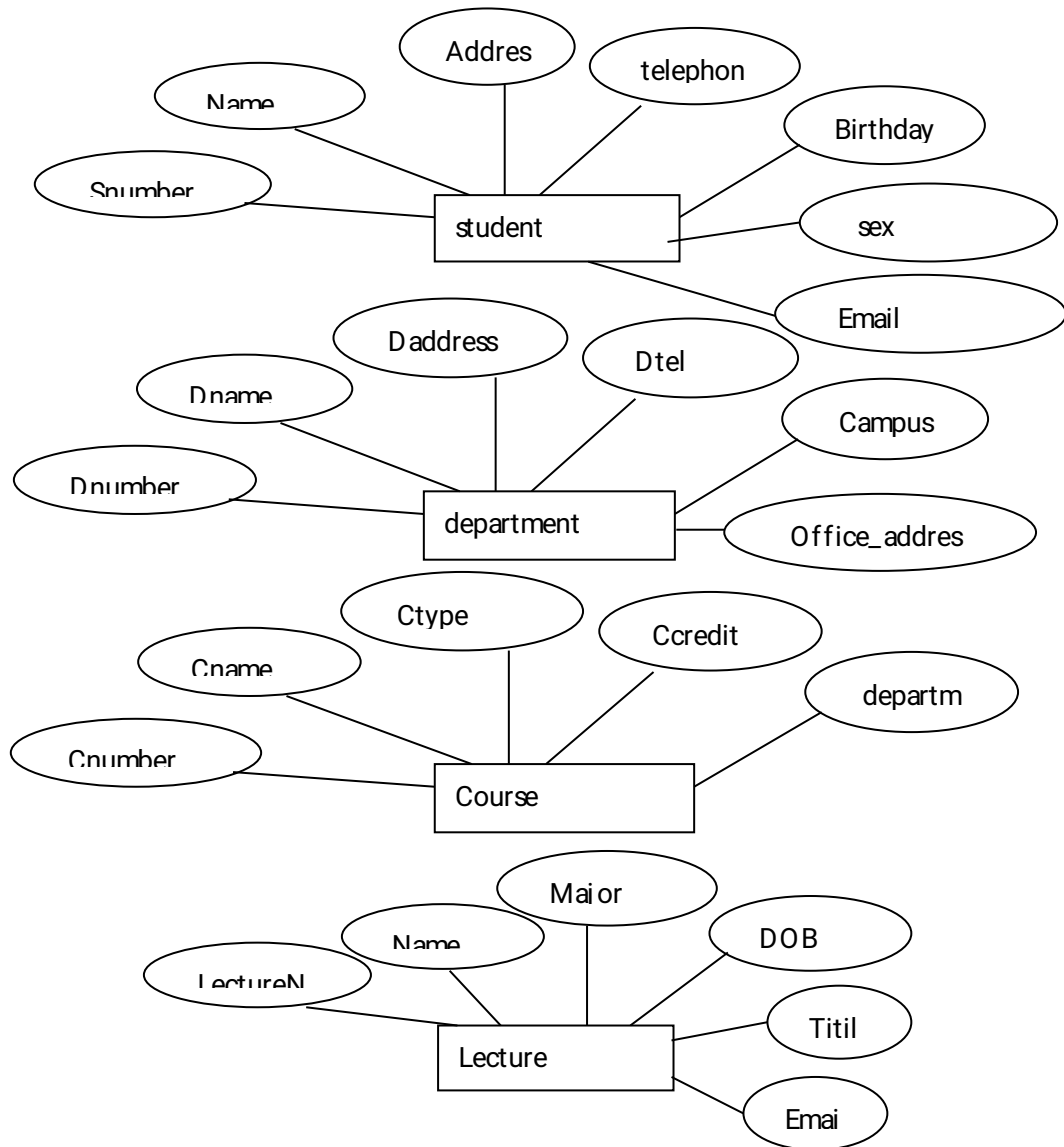
Suppose we are given relation R and S as below, please give results of the following relational algebra expressions

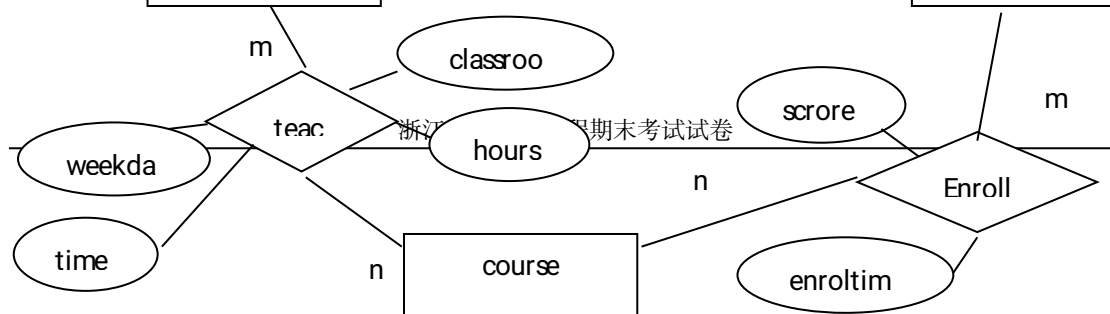
T			U		V	
A	B	C	B	D	C	E
1	2	3	1	3	3	1
1	1	3	2	4	3	2
2	1	4	1	5	4	6
2	2	4	2	6	4	7
3	4	5	1	9		
3	1	4	2	7		

1) $\Pi_B(T) \cap \Pi_B(U)$

- 2) $\prod_c(T) \cup \prod_c(V)$
- 3) $\prod_c(T) - \prod_c(V)$
- 4) $\sigma_{T.C=3} (T \bowtie U)$
- 5) $\sigma_{T.C=V.C=3} (T \times V)$

5. Map the following ER diagram into relation schemas (point:10)





6. (point: 13) Consider the following relations for sale activities

Sale(Cno, Cname, Caddress, Ctel, Pno, Pname, price, Pcategory, Warehouse, Quantity)

Cno, Cname, Caddress, Ctel represent the number, name, address, telephone number of the customers, **Pno, Pname, price, Pcategory** represent the number, name, price, category(种类) of the products sale to the customers, for each category of product a certain warehouse is specified. We also record the quantity when product is sold. One customer can buy one or more than one products and one products can be purchased by one or more than one customers.

- Identify the primary key of this table; (point:2)
- Identify all the functional dependencies within the above relation; (point: 3)
- What normal form is the relation in? Explain your answer.(point: 2)
- Apply normalization until the relations are BCNF. State the reasons behind each decomposition. (point:6)

7. (point: 32) The following tables form part of a database held in a relational DBMS.

Driver (Dnumber, Dname, Dsex, Dage, Daddress, Spv_Dnum)

Bus (Bnumber, Btype, capacity)

Schedule (RouteNo, Bnumber, Dnumber, date, d_time)

Each driver has a unique number represented by '**Dnumber**'. The name, sex, age, address of the driver represented by **Dname, Dsex, Dage, Daddress** respectively, and the **Spv_Dnum** is the number of the supervisor. Every bus has a unique number (Bnumber) and a type (Btype) and a capacity (=maximum number of passengers, 车子的座位数). Furthermore, we have a table that contains the scheduling of buses and drivers on existing routes (公交线路). All routes start at the central bus station. The attribute 'time' indicates the departure time (发车时间). One route has a number of buses and drivers, and for each day, a bus and a driver can only be scheduled on one route.

(1) Identify the primary keys, candidate keys (if have) and foreign keys of the above relations (point: 6)

(2) Formulate the following queries in RA and SQL respectively: (point:16)

- List the numbers, types of the buses with a capacity of more than 60.
- Give the numbers, names of the drivers who are scheduled today (getdate()).
- List the numbers of the buses which is not scheduled today.
- Give the numbers and names of the drivers who have driven on all routes.

(3) Formulate the following questions in SQL: (point:10)

- Give the total number of drivers of the company.
- Give the number of buses of on each route today.

Q 7: create a view which contains the buses numbers which have worked more than 25 days this month.

Q 8: Give the names of the bus drivers who are driving on more than 3 routes.

Q 9: Who has never driven a bus with a capacity of more than 80?